SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE NAME: Artificial Intelligence **COURSE CODE: CSA17**

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. *(BL3)*

Assessment Tool Weightage: 30%

Duration : 45 Minutes

QUESTIONS: 2 **Total Marks:20**

Annexure A

Descriptive Written Test

Code of Conduct:

I Praveen J (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Annexure A

Descriptive Written Test

1. A smart home automation system is being designed to manage lighting, temperature, and security based on user behavior and environmental conditions. The system must respond to real-time inputs such as motion detection, temperature changes, and time of day while also learning user preferences over time. In this context, explain the different types of intelligent agents (simple reflex, model-based, goal-based, and utility-based agents) and analyze how each type would function within this system. Justify which type of agent or hybrid approach would be most effective for ensuring comfort, energy efficiency, and security.

1. Smart home automation agents

An intelligent agent perceives its environment through sensors, chooses actions, and acts through devices such as lights, thermostats, and locks. The agent types differ in how they choose those actions:

- **Simple reflex agent:** Responds directly to the current input using condition–action rules. For example, "if motion is detected and it is dark, turn on the lights." It is fast, but does not remember earlier events or consider long-term effects.
- **Model-based agent:** Keeps an internal model of the home, including relevant past observations and current estimates. It can, for instance, remember that a room is occupied even after motion stops briefly. This helps when the environment is only partly observable.
- **Goal-based agent:** Chooses actions that help achieve a stated goal, such as maintaining a target temperature or ensuring the house is locked at night. It can compare possible actions, but a goal alone may not express which outcome is preferable when goals conflict.
- **Utility-based agent:** Ranks possible outcomes by their usefulness. It can balance comfort, energy use, and security—for example, choosing a slightly less comfortable temperature setting to save energy when nobody is home.

For this system, a **hybrid agent** is most effective. Simple reflex rules can handle urgent events, such as turning on lights when someone enters or triggering a security response. A model-based component can track occupancy and learn preferences over time. A utility-based component can balance comfort and energy efficiency when selecting temperature and lighting settings. This combination supports quick responses while also adapting to context and competing priorities.

2. A robot is placed in an unknown maze and must find a path to the exit without prior knowledge of the layout. Explain how uninformed search strategies like Uniform Cost Search (UCS) and Iterative Deepening Search (IDS) can help solve this problem. Discuss the advantages and disadvantages of these algorithms in terms of time and space complexity. Relate the problem to different types of intelligent agents and explain how agent design affects decision-making. Suggest which combination of agent type and search strategy would be most effective and justify your choice.

2. Finding an exit in an unknown maze

A robot with no prior map can explore the maze by treating locations as states and possible moves as actions. As it moves, it records what it has discovered. **UCS** and **IDS** are uninformed search strategies: they do not use a heuristic estimate of how close a location is to the exit.

Uniform Cost Search (UCS)

UCS expands the least-cost path found so far. It uses a priority queue ordered by accumulated path cost.

- **Advantage:** It finds an optimal path when action costs are nonnegative.
- **Disadvantage:** It may use substantial time and memory because it stores many frontier paths.
- **Complexity:** With branching factor \(b\), minimum step cost \(\epsilon>0\), and optimal solution cost \(C^*\), its time and space can be exponential in \(C^*/\epsilon\).

UCS is useful when maze moves have different costs, such as when some routes are longer or more dangerous.

Iterative Deepening Search (IDS)

IDS repeatedly performs depth-limited search, increasing the depth limit each time until it finds the exit.

- **Advantage:** It uses much less memory than UCS, retaining mainly the current search path.
- **Disadvantage:** It revisits shallow locations in successive iterations, and it does not generally find the least-cost path when step costs vary.
- **Complexity:** For unit-cost moves, its time is \(O(b^d)\), where \(d\) is the depth of the shallowest exit; its space is \(O(bd)\).

IDS is a good fit when each move has the same cost and memory is limited.

Agent design and recommendation

A **model-based learning agent** is suitable for the robot. Its sensors provide information about nearby passages, its internal map records explored and blocked locations, and its decision-making component selects the next move. It should update the map after each observation and replan when it finds a blocked passage or new route.

If every move has the same cost and the robot has limited memory, combine this agent design with **IDS**. If movement costs differ and finding the least-cost route matters, use **UCS** instead, accepting its higher memory use. The search method determines how routes are compared; the agent design determines how the robot senses, stores what it learns, and responds to new information.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Criteria	**Level 4 (Exemplary)**	**Level 3 (Proficient)**	**Level 2 (Developing)**	**Level 1 (Beginning)**	**Max Marks**
Understanding of Concepts	25%	Demonstrates thorough, accurate understanding of concepts	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor understanding; major misconceptions
Explanation	25%	Detailed, well-explained answers with relevant examples	Clear explanation with adequate details	Limited explanation; lacks depth	Poor or unclear explanation
Application	20%	Effectively applies concepts with strong analytical ability	Applies concepts with minor errors	Limited application with weak analysis	No application or incorrect analysis
Organization	15%	Well-structured answers with logical flow and coherence	Organized with minor issues in flow	Basic structure, lacks clarity	Poor organization, incoherent answers
Accuracy	10%	Completely accurate and covers all required points	Mostly accurate with minor omissions	Partially correct with missing points	Incorrect answers with major gaps
Clarity	5%	Clear, neat, professional presentation	Understandable with minor issues	Limited clarity, readability issues	Poorly written, difficult to understand